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Tailored design of Audio–Visual Speech Recognition models using Branchformers

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ABSTRACT

Recent advances in Audio–Visual Speech Recognition (AVSR) have led to unprecedented achievements in the field, improving the robustness of this type of system in adverse, noisy environments. In most cases, this task has been addressed through the design of models composed of two independent encoders, each dedicated to a specific modality. However, while recent works have explored unified audio–visual encoders, determining the optimal cross-modal architecture remains an ongoing challenge. Furthermore, such approaches often rely on models comprising vast amounts of parameters and high computational cost training processes. In this paper, we aim to bridge this research gap by introducing a novel audio–visual framework. Our proposed method constitutes, to the best of our knowledge, the first attempt to harness the flexibility and interpretability offered by encoder architectures, such as the Branchformer, in the design of parameter-efficient AVSR systems. To be more precise, the proposed framework consists of two steps: first, estimating audio- and video-only systems, and then designing a tailored audio–visual unified encoder based on the layer-level branch scores provided by the modality-specific models. Extensive experiments on English and Spanish AVSR benchmarks covering multiple data conditions and scenarios demonstrated the effectiveness of our proposed method. Even when trained on a moderate scale of data, our models achieve competitive word error rates (WER) of approximately 2.5% for English and surpass existing approaches for Spanish, establishing a new benchmark with an average WER of around 9.1%. These results reflect how our tailored AVSR system is able to reach state-of-the-art recognition rates while significantly reducing the model complexity w.r.t. the prevalent approach in the field. Code and pre-trained models are available at https://github.com/david-gimeno/tailored-avsr.

1. Introduction

Automatic Speech Recognition (ASR) is a widely studied task in the field of speech processing (Gales and Young, 2008). Albeit current ASR systems are nowadays capable of understanding spoken language with great quality (Radford et al., 2023; Baevski et al., 2020), their performance can be considerably affected in adverse scenarios, such as a noisy environment (Juang, 1991). This challenge has led to the emergence of Visual Speech Recognition (VSR) technologies as an interesting avenue of research (Fernandez-Lopez and Sukno, 2018; Ma et al., 2022; Chang et al., 2024). Nonetheless, inspired by our speech perception process (McGurk and MacDonald, 1976; Besle et al., 2004; Stacey et al., 2020), researchers have increasingly turned their focus towards Audio–Visual Speech Recognition (AVSR) as a promising solution, demonstrating how the incorporation of visual cues when interpreting speech

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improves the performance of this type of systems in challenging, noisy scenarios (Afouras et al., 2018a; Ma et al., 2021b; Shi et al., 2022; Burchi and Timofte, 2023; Ma et al., 2023; Fernandez-Lopez et al., 2024). While multiple methods have been investigated to effectively integrate both modalities (Potamianos et al., 2003; Sterpu et al., 2018; Wei et al., 2020; Wu et al., 2021), the prevalent AVSR architecture nowadays is based on early fusion strategies, either combining the speech features extracted by two independent encoders (Afouras et al., 2018a; Ma et al., 2021b, 2023; Burchi and Timofte, 2023), each dedicated to a specific modality, or adopting the recent trend of unified multi-modal encoders (Shi et al., 2022; Zhu et al., 2023; Li et al., 2023).

However, although these approaches represent the current state of the art, determining the optimal cross-modal architectures remains an ongoing challenge. A significant limitation of existing studies lies not only in their reliance on models with vast amounts of parameters and extensive pretraining processes, but also in their focus on how audio–visual features should be fused to enhance the performance of a fixed encoder backbone. Therefore, it is not clear how encoder architectures can be adapted to the different nature and intrinsic characteristics of each modality. The Branchformer encoder (Peng et al., 2022; Kim et al., 2023a) seemed to represent one step forward in this regard. However, apart from being proposed solely in the context of auditory-based ASR, the authors did not exploit the potential of the flexibility and interpretability offered by this architecture for the design of tailored and parameter-efficient systems. These were the reasons that motivated our work, whose key contributions are:

- We propose a novel framework to design tailored, unified audio–visual encoders capable of reaching state-of-the-art performance, while reducing the number of model parameters by nearly half, e.g., from 103.5M to 59.3M in the model trained for English, when compared to the conventional and prevalent AVSR architecture.
- To the best of our knowledge, our work is one of the first attempts to harness the flexibility and interpretability offered by encoder architectures, such as the Branchformer, in the design of parameter-efficient AVSR systems. Specifically, we leverage these properties to identify the context dependencies – whether short- or long-term – that are more relevant for processing each modality. This analysis directly informs the design of a tailored AVSR system, optimized to handle the distinct characteristics of audio and visual inputs effectively.
- Through extensive experiments on two English (LRS2-BBC (Afouras et al., 2018a) and LRS3-TED (Afouras et al., 2018b)) and four Spanish (MuAViC (Anwar et al., 2023), CMU-MOSEAS (Zadeh et al., 2020), LIP-RTVE (Gimeno-Gómez and Martínez-Hinarejos, 2022), and VLRF (Fernandez-Lopez et al., 2017)) AVSR benchmarks, we demonstrated the effectiveness of our proposed audio–visual framework across diverse data conditions and scenarios. Even with a moderate scale of training data and a significantly reduced number of model parameters, our models remarkably achieve competitive word error rates (WER) of approximately 2.5%, on par with state-of-the-art methods (Ma et al., 2021b; Burchi and Timofte, 2023) under similar conditions. For Spanish datasets, the framework surpasses existing approaches, achieving an average WER around 9.1%, thereby establishing a new benchmark for this language.
- Our analysis provides initial insights into the architectural differences, beyond the frontends, in processing speech features across modalities. Acoustic cues rely on both short- and long-term relationships, whereas the video-only system heavily depends on global-context cues, likely reflecting the ambiguity and limited information in lipreading. Additionally, our adaptive audio–visual fusion module underscored the dominant role of acoustic cues, with a relevance of approximately 70% — a finding that aligns with studies indicating that only 30% of speech information is visible (Duchnowski et al., 2000).

2. Related work

State of the Art in AVSR. Thanks to the collection of large-scale datasets (Afouras et al., 2018b; Anwar et al., 2023) and the use of powerful attention-based mechanisms (Vaswani et al., 2017), unprecedented results have been achieved in the context of AVSR (Shi et al., 2022; Burchi and Timofte, 2023; Ma et al., 2023; Chang et al., 2024). While most of these studies are predominantly evaluated on English — with performances between 0.8–0.9%, 1.0–1.5%, and 12%–20% WER for AVSR, ASR, and VSR tasks, respectively (Ma et al., 2023; Chang et al., 2024) — the performance for other languages, such as Spanish, remains comparably distant. Though recent works are considering more languages in their studies (Ma et al., 2022; Kim et al., 2023b; Anwar et al., 2023; Hong et al., 2023), Spanish AVSR performance is notably lower with an average WER of approximately 16%, 15%, 50% WER for AVSR, ASR, and VSR tasks, respectively. In terms of model architecture, the hybrid CTC/Attention framework (Watanabe et al., 2017) is the most prevalent decoding paradigm adopted in the field. In our research, we also adopt this architecture, aiming to assess the effectiveness of our proposed audio–visual encoder more compellingly.

Parameter-Efficient ASR Models. Several works have focused their efforts on building parameter-efficient ASR models through diverse methods, including distillation (Li et al., 2018), quantization (Nguyen et al., 2020; Sainath et al., 2020), pruning (Lai et al., 2021; Ding et al., 2022), and parameter-sharing techniques (Wei et al., 2023). Compatible with structured sparsity, Ding et al. (2022) investigated the search of optimal sparse subnetworks within audio-only ASR architectures, not only without experiencing a significant drop in performance but also achieving robustness under noisy conditions. Although limited, similar methods were also applied for VSR, highlighting Fernandez-Lopez et al. (2023), which explored multiple pruning strategies, also demonstrating the potential robustness of these sparse models to noise compared to their dense counterparts. For AVSR, the increasing computational demands have led to innovations such as mask optimization training procedures aimed at improving scalability (Fernandez-Lopez et al., 2024). Of particular note, however, is the work by Burchi and Timofte (2023), who introduced a non-autoregressive CTC-based AVSR architecture. Their approach not only achieved competitive results but also reduced model complexity through an

efficient attention mechanism. However, unlike our research, their proposed model offered no architectural innovation in the fusion of audio–visual speech cues.

Interpreting ASR. Multiple studies demonstrated that as we progress through the encoder layers, low-level patterns like speaker characteristics and noise are gradually disregarded, leading to a more standardized representation of high-level semantics (Mohamed et al., 2012; Li et al., 2020). The primary focus of other works has been to interpret the underlying phonetic representations captured by these deep neural networks. This was achieved either through phonetic-based regularization techniques during training (Tan et al., 2015; Wu et al., 2018) or the incorporation of additional modules to provide more explainable speech representations, with the goal of identifying the most relevant features for the model (Ravanelli and Bengio, 2018; Agrawal and Ganapathy, 2020). While these studies focused on improving the interpretation of ASR systems, the Branchformer encoder (Peng et al., 2022) represents one of the first attempts in addressing the problem from the perspective of model architecture design. As described in Section 3.1, this encoder allows researchers to interpret whether the global or local dependencies are more relevant at each encoder layer when processing speech. However, despite conducting several interesting studies, the authors did not exploit the potential of this novel approach towards the design of optimal and parameter-efficient ASR architectures as we propose in this paper. In addition, unlike our work, all these studies were focused on auditory-only ASR without considering the audio–visual integration and how both modalities differentiate and complement each other.

Cross-Modal Unified Encoders for AVSR. Shi et al. (2022) introduced AV-HuBERT, a self-supervised learning framework for AVSR capable of reaching state-of-the-art results on the LRS3-TED benchmark (Afouras et al., 2018b). The audio- and video-based cues are first processed by modality-independent frontends. These frontend features are then fused and fed into a shared backbone Transformer estimated with a frame-level masked cluster prediction task. Zhu et al. (2023) presented VATLM, an extension of the AV-HuBERT framework that not only integrates audio and visual cues, but it also incorporates the corresponding text transcription as an additional input to the unified cross-modal encoder backbone. Both architectures presented a base and large variant comprising around 105M and 330M, respectively. However, the substantial performance gap between the base and large models in both works reflects how the effectiveness of these architectures highly depends on the number of parameters that they comprise. Similar findings are found for the cross-lingual XLAVS-R model (Han et al., 2024), whose architecture is mainly based on AV-HuBERT. Conversely, Li et al. (2023) compared multiple methods to fuse the audio and visual cues using a supervised AVSR architecture without relying on large-scale pretraining processes. Their experiments showed superior performance when concatenating both modalities along the temporal dimension, thus offering a robust approach to noisy scenarios that does not require forced alignment despite the modality mismatch in terms of sampling rate.

Our present work. However, these studies were limited to how the audio–visual features should be fused to maximize the performance of a fixed unified encoder, which was not modified or adapted to the different nature and intrinsic characteristics of each modality. Unlike prior research, our proposed method focused on the tailored design of the optimal cross-modal encoder architecture for parameter-efficiency AVSR by leveraging the interpretability and flexibility of the Branchformer encoder (Peng et al., 2022). Furthermore, we show the effectiveness of our proposed method through extensive experiments in two different languages and a wide range of data conditions, even when estimated with a moderate amount of training data.

3. Method

For clarity, we first describe the conventional and prevalent end-to-end architecture for AVSR in Section 3.1, as background, to subsequently introduce our proposed tailored and unified audio–visual encoder in Section 3.2.

3.1. Model architecture

Audio Frontend. The audio frontend transforms the raw audio waveforms into 80 mel-spectrogram coefficients by applying a short-time Fourier transform with a 20 ms window and a 10 ms step size. Then, a two-layer convolutional module is not only in charge of extracting more powerful local and temporal acoustic features, but also of down-sampling the pre-processed audio sequence to a quarter of its length. This down-sampling addresses the mismatch in sample rates between audio and visual cues, which typically correspond to 100 fps and 25 fps, respectively, as it is the case with the datasets in our case study. Consequently, audio and visual cues are temporally aligned. The final acoustic representation sequence consists of 256-dimensional embeddings.

Visual Frontend. Consistent with prior works (Ma et al., 2022, 2023), spatio-temporal relationships are captured by using a 3D convolutional layer with a kernel of 7×7 pixels and a receptive field of 5 frames. Once these visual features are flattened along the time dimension, a 2D ResNet-18 (He et al., 2016) is responsible for extracting local visual patterns, resulting in a 512-dimensional feature representation.

Branchformer Encoder. Both the acoustic and visual feature sequences are projected to a 256-dimensional space and further injected with a relative positional encoding (Dai et al., 2019). Then, these enhanced frontend representations are each passed through a different modality-specific encoder inspired by the Branchformer architecture (Peng et al., 2022) and its enhanced variant, E-Branchformer (Kim et al., 2023a). This enhanced variant not only explored more advanced convolution-based merging techniques, but also incorporated feed-forward modules in a macaron-style configuration. Each of the 12 layers in our encoder consists primarily of two parallel branches:

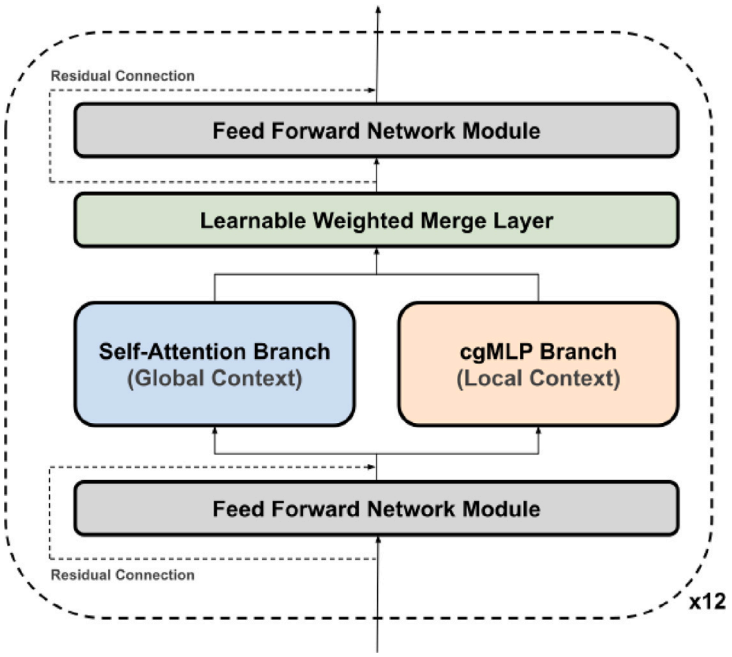


Fig. 1. Encoder layer design inspired by the Branchformer architecture (Peng et al., 2022; Kim et al., 2023a).

- **Self-Attention Branch.** This branch is responsible for modeling the global long-term dependencies from speech cues using a multi-headed self-attention mechanism (Vaswani et al., 2017). Concretely, we set to 4 and 64 the number and size of the attention heads comprising the module, respectively.
- **cgMLP Branch.** A Multi-Layer Perceptron with Convolutional Gating (cgMLP) module (Sakuma et al., 2022) is designed to capture the local context relationships. Specifically, the input sequence is first projected to an up-sampling inner dimension of 2048. Then, the sequence is split to process part of it using a 1D depth-wise convolutional layer with a kernel size of 31 to subsequently apply a linear gating mechanism. Finally, a linear layer projects the obtained representation to its original dimension of 256.

As shown in the scheme depicted in Fig. 1, the encoder layers also incorporate two feed-forward modules with an up-sampling inner dimension of 2048 set in a macaron-style manner alongside their corresponding residual connections with a scale factor of 1/2. In all cases, the Swish activation function (Ramachandran et al., 2017) is used, layer normalization (Ba et al., 2016) precedes each of the described modules, while a final dropout (Srivastava et al., 2014) of 0.1 is applied over their output. Finally, the latent representations provided by both branches are merged via a learnable weighted method that, as described below (and reflected in Fig. 2), allowed us to interpret which branch was more relevant at each encoder layer for the audio- and video-only case studies.

Learnable Weighted Merge Layer. This module is in charge of combining the output of both branches and dynamically generating the weights representing the relevance of each branch at each encoder layer through an attention-based pooling projection process. Let $\boldsymbol{Z} \in \mathbb{R}^{T \times d}$ be the d-dimensional audio or video sequence of length T processed by one of the encoder branches and $\boldsymbol{w} \in \mathbb{R}^{d}$ be a learnable parameter vector. The attention weights $\boldsymbol{\alpha} \in \mathbb{R}^{T}$ for each vector $\boldsymbol{z}_i \in \mathbb{R}^{d}, 1 \leq i \leq T$ composing $\boldsymbol{Z}$ are computed as:

$$\alpha_i = \frac{\exp\left(\boldsymbol{w}^T \boldsymbol{z}_i / \sqrt{d}\right)}{\Sigma_{j=1}^{T} \exp\left(\boldsymbol{w}^T \boldsymbol{z}_j / \sqrt{d}\right)}, \tag{1}$$

representing the scaled dot product between $\boldsymbol{Z}$ and $\boldsymbol{w}$ followed by a softmax normalization. Once the attention scores $\boldsymbol{\alpha}$ were computed, we transform the sequence $\boldsymbol{Z}$ into the output $\boldsymbol{a} \in \mathbb{R}^{d}$ via a weighted sum: $\boldsymbol{a} = \Sigma_{i=1}^{T} \alpha_i \boldsymbol{z}_i$. Therefore, based on this procedure, we could transform the output sequences from the self-attention and cgMLP branches into the attention-enriched representations denoted by $\boldsymbol{a}_{att}$ and $\boldsymbol{a}_{mlp}$, respectively. Then, these two vectors are projected to two scalars and normalized using softmax to obtain the branch relative weights as follows:

$$w_{attn}, w_{mlp} = softmax\left(\boldsymbol{W}_{attn} \boldsymbol{a}_{attn}, \boldsymbol{W}_{mlp} \boldsymbol{a}_{mlp}\right), \tag{2}$$

where $\boldsymbol{W}_{att}, \boldsymbol{W}_{mlp} \in \mathbb{R}^{1 \times d}$ are linear transforms. Finally, the encoder provides a merged output $\boldsymbol{X} \in \mathbb{R}^{T \times d}$ capturing both global and local speech patterns computed as defined in:

$$\boldsymbol{X} = w_{att} \boldsymbol{Z}_{att} + w_{mlp} \boldsymbol{Z}_{mlp}, \tag{3}$$

where $\boldsymbol{Z}_{att}$ and $\boldsymbol{Z}_{mlp}$ refer to the output latent representations provided by the self-attention and cgMLP encoder branches, respectively. In this work, d corresponds to 256 dimensions.

Adaptive Audio–Visual Fusion. We present an adaptive audio–visual fusion module based on the learnable weighted merge layer described above. Let $\boldsymbol{X}_{audio}$ and $\boldsymbol{X}_{video}$ be the audio and video sequences encoded by the corresponding modality-specific encoder, respectively, the audio–visual speech representation is defined such that:

$$\boldsymbol{X} = PoswiseFFN\left(w_{audio}\boldsymbol{X}_{audio} + w_{video}\boldsymbol{X}_{video}\right), \tag{4}$$

where w_{audio} and w_{video} are the modality relative weights resulting from an attention-based pooling projection process, and $PoswiseFFN$ represents a 2-layer position-wise feed-forward module with an inner and output dimension of 2048 and 256, a Swish activation layer, and a dropout rate of 0.1.

CTC/Attention Decoder. This state-of-the-art hybrid architecture (Watanabe et al., 2017) combines both the Markov assumptions of CTC (Graves et al., 2006) (properties in harmony with the speech nature) and the flexibility of the non-sequential alignments offered by the attention-based decoder (Vaswani et al., 2017). As Fig. 3 depicts, this module is composed of a linear projection as the CTC-based decoding branch and a 6-layer Transformer decoder of 4 attention heads of 64 dimensions attached to the Cross Entropy (CE) criterion. As proposed by Watanabe et al. (2017), its loss function is defined as follows:

$$\mathcal{L} = \alpha \log p_{ctc}(\boldsymbol{Y}|\boldsymbol{X}) + (1-\alpha)\log p_{attn}(\boldsymbol{Y}|\boldsymbol{X}), \tag{5}$$

where p_{ctc} and p_{attn} denote the CTC and Attention posteriors, respectively. In both terms, $\boldsymbol{X}$ and $\boldsymbol{Y}$ refer to the speech latent representation processed by the encoder and its corresponding character-level target transcription, respectively. The α weight balances the relative influence of each decoding branch.

Language Model. In this work, we use the character-level Language Models (LM) publicly released by Ma et al. (2022) for Spanish and English. These models, defined as 16-layer Transformer encoders (Vaswani et al., 2017), were estimated with text from different sources comprising more than 150M characters each.

Inference. In the CTC/Attention architecture inference process, the Attention-based decoder primarily determines when the beam search decoding concludes by predicting the end-of-sentence token, and thus both the CTC-based decoding branch and the LM serve as additional scoring factors, as reflected by

$$S = \lambda S_{ctc} + (1-\lambda)S_{attn} + \beta S_{lm}, \tag{6}$$

where S_{ctc} and S_{attn} are the scores of the CTC- and the Attention-based decoder, respectively, λ is their corresponding relative weight, and β and S_{lm} refer to the LM decoding influence weight and the LM score, respectively. Readers are referred to Watanabe et al. (2017) for a more comprehensive description of this inference process.

3.2. The tailored audio–visual branchformer

The model architecture previously described, composed of two independent modality-specific encoders, corresponds to the most prevalent approach adopted in the context of AVSR (Afouras et al., 2018a; Ma et al., 2021b; Prajwal et al., 2022; Ma et al., 2023; Burchi and Timofte, 2023). While recent works have explored unified encoder architectures (Shi et al., 2022; Zhu et al., 2023; Li et al., 2023), determining the optimal cross-modal architecture remains an ongoing challenge. In this paper, unlike prior works, we introduce a novel framework that, to the best of our knowledge, constitutes the first attempt to harness the flexibility and interpretability offered by encoder architectures, such as the Branchformer (Peng et al., 2022), in the design of parameter-efficient AVSR systems. The proposed method comprises multiple steps.

Training Modality-Specific Models. We first estimated models for audio-only and video-only inputs separately, each utilizing its corresponding modality-specific Branchformer encoder: one dedicated to the processing of audio speech cues (E_{audio}) and the other to visual speech cues (E_{video}).

Analyzing Branchformer Encoder Layers. As described in Section 3.1, the Branchformer encoder allowed us to analyze which branch was more relevant at each encoder layer for each of our estimated modality-specific models. Fig. 2 reflects how the encoder adopted a different architecture depending on the addressed modality for the LRS2-BBC dataset. The increased reliance on global context relationships in the video-only scenario, compared to the audio-only setting, could be attributed to the limited information provided by lip movements when interpreting speech (Duchnowski et al., 2000). Section 5 discusses the patterns observed across the datasets explored in our work.

Designing a Tailored AVSR Model. We also observed that, in most cases and regardless of the modality, one of the relative branch weights of the modality-specific models tended to approach saturation, with values consistently close to 1. Our hypothesis was that, due to this saturation, the branch associated with the lower impact could be removed without a significant decrease in model performance. Based on this analysis, we designed a tailored, unified, parameter-efficient encoder for AVSR, whose overview scheme is depicted in Fig. 3. This tailored design involves the selection of ad-hoc global- or local-context processing modules depending on the modality and the specific encoder layer level we are addressing. Specifically, our proposed tailored encoder is composed of 12 layers with two parallel branches but, in this case, each one is in charge of processing a different modality. Each layer of

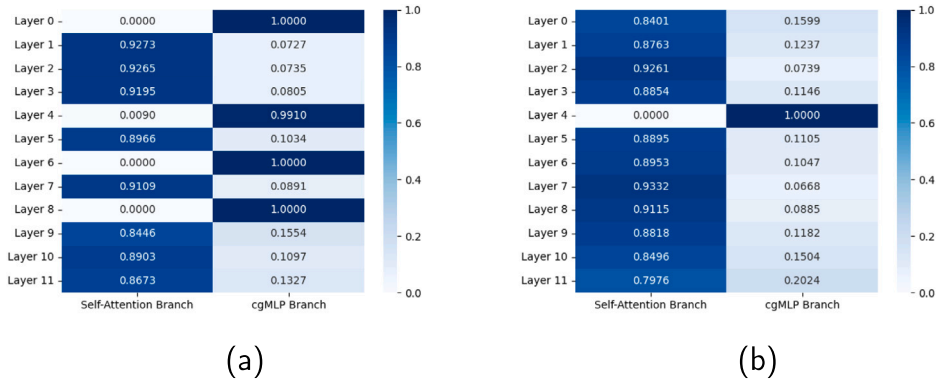


Fig. 2. Visualization of the average Branchformer weights on the LRS2-BBC validation data set. (a) Acoustic Speech Recognition. (b) Visual Speech Recognition.

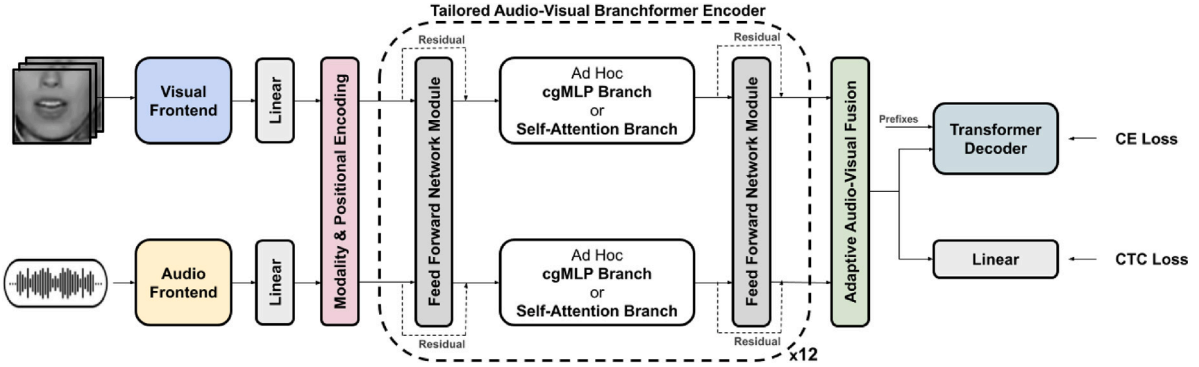


Fig. 3. Our proposed method involves preprocessing visual and audio cues, conditioning the resulting temporal-aligned speech features with positional and modality embeddings, and then modeling them using a tailored encoder, whose architecture design is based on the interpretable weights of models previously estimated for audio- and video-only settings, to perform the final speech interpretation in an autoregressive manner through the hybrid CTC/Attention paradigm. Both feed forward networks share parameters across modalities. CE and CTC refer to Cross Entropy and Connectionist Temporal Classification, respectively.

these branches is defined by a self-attention or cgMLP module depending on which of them was more relevant in its corresponding modality-specific model. Formally, this can be expressed as follows:

$$\delta_{im}(x_m) = \begin{cases} cgMLP(x_m) & \text{if } E^{global}_{im}(x_m) < E^{local}_{im}(x_m), \\ Self\,Attention(x_m) & \text{otherwise,} \end{cases} \tag{7}$$

where i denotes the layer index, $m \in \{audio, video\}$ indicates the modality, and x_m represents the input features for modality m. Therefore, δ_{im} corresponds to the context-specific processing module of the ith layer of the tailored, unified audio–visual encoder. Additionally, $E^{global}_{im}(x_m)$ and $E^{local}_{im}(x_m)$ refer to the scores provided by our modality-specific models that indicate the relevance of global and local context at layer i for modality m, respectively.

Unlike the original Branchformer encoder, we do not merge the encoded representation of each branch. Doing so would fundamentally alter the nature of the audio- and video-only representations, thus hindering the ad-hoc design described above by not having a base model to select one module focused on either global or local context relationships. Nonetheless, we share the two feed-forward modules with the intention of enhancing the model's cross-modal capabilities. Consequently, in order to facilitate the model to distinguish between both modalities, we introduced a modality embedding that conditions the frontend speech representations. A preliminary ablation study showed that, although no significant differences were observed, incorporating the modality embedding consistently led to slight improvements.

As supported by our extensive experiments, this tailored architecture design allowed us to reduce the number of model parameters by nearly half compared to conventional AVSR systems, while achieving state-of-the-art recognition rates across diverse benchmarks.

4. Experiments

The models proposed in this work were implemented using the open-source ESPNet toolkit (Watanabe et al., 2018). Experiments were conducted on a 24-core 3.40 MHz Intel i7-13700K CPU and a GeForce RTX 4090 GPU with 24 GB memory.

4.1. Databases

In our work, we considered multiple AVSR benchmarks covering two different languages, as well as a wide range of scenarios and data conditions, to properly evaluate the effectiveness of our proposed audio–visual framework.

LRS2-BBC (Afouras et al., 2018a) is a large-scale English database composed of around 224 h collected from BBC TV programs. It consists of a pre-training set with 96,318 samples (195 h), a training set with 45,839 samples (28 h), a validation set with 1082 samples (0.6 h), and a test set with 1243 samples (0.5 h). It offers over 2M running words with a vocabulary size of around 40k.

LRS3-TED. (Afouras et al., 2018b) is the largest publicly audio–visual English database, providing around 438 h collected from TED talks. It consists of a 'pre-train' set with 118,516 samples (407 h), a 'train-val' set with 31,982 samples (30 h), and a test set with 1321 samples (0.9 h). It offers more than 4M running words with a vocabulary size of around 50k.

MuAViC$_{es}$ (Anwar et al., 2023) is a large-scale multilingual corpus collected from TED and TEDx talks (Salesky et al., 2021). In our work, we only considered its Spanish partition, offering around 181 h of data. It is split into a training set with 102,035 samples (177.5 h), a validation set with 906 samples (1.6 h), and a test set of 1012 samples (1.7 h). It comprises around 1.7M running words with a vocabulary size of around 58k.

CMU-MOSEAS$_{es}$ (Zadeh et al., 2020) is a large-scale database collected from YouTube monologue videos covering 4 different languages. In our work, we only considered its Spanish partition. According to Ma et al. (2022), we defined a training set with 8253 samples (15.7 h) and a test set with 329 samples (0.6 h). The database provides more than 150k running words with a vocabulary size of around 14k.

LIP-RTVE (Gimeno-Gómez and Martínez-Hinarejos, 2022) is a challenging Spanish database collected from TV newscast programs, providing around 13 h of data. Its speaker-independent partition consists of a training set with 7142 samples (9 h), a validation set with 1638 samples (2 h), and a test set with 1572 samples (2 h). It comprises around 300 speakers and more than 100k running words with a vocabulary size of around 10k.

VLRF (Fernandez-Lopez et al., 2017) is a Spanish database primarily designed to study the feasibility of the VSR task, providing 1 h of data. It was recorded in controlled laboratory settings, providing a speaker-dependent partition composed of 24 speakers, each one with 25 assigned unrelated but predefined sentences. It is split into a training set with 480 samples (0.92 h) and a test set with 120 samples (0.23 h), comprising around 4k running words with a vocabulary size of 1374.

4.2. Model architecture comparison

In our work, we compared three different types of models.

Audio- and Video-Only. These modality-specific systems constitute the baseline from which we will extract the Branchformer-based knowledge to design our proposed tailored AVSR system. Note that in these audio- and video-only settings, the model architecture described in Section 3.1 is modified so that, in addition to removing the audio–visual fusion module, only the corresponding frontend is considered. The audio- and video-only models comprise around 51.2M and 60.7M parameters, respectively, for both languages.

Conventional AVSR. This model corresponds to the architecture described in Section 3.1, where a modality-specific Branchformer encoder is defined for each modality. This model represents the conventional architecture adopted in the context of AVSR, comprising around 103.5M parameters for both languages studied in our work.

Tailored AVSR. This model corresponds to the architecture described in Section 3.2, where a tailored audio–visual Branchformer was designed. This model represents a step forward in parameter efficiency in the context of audio–visual speech integration, comprising around 59.3M or 58.3M parameters for English and Spanish, respectively.

4.3. Implementation details

Audio Processing. We first sampled all audio waveforms at 16 kHz and then applied utterance-level mean subtraction.

Video Processing. All videos were first sampled at 25 fps. Similar to Burchi and Timofte (2023) and Ma et al. (2021b, 2022), we cropped patches of 96×96 pixels centered on the speaker's mouth. We used the RetinaFace face detector (Deng et al., 2020) and the Face Alignment Network (Bulat and Tzimiropoulos, 2017) to detect 68 facial landmarks. Subsequently, we applied a similarity transformation w.r.t. a neutral reference frame, thereby removing possible translation and scaling variations. The cropped lip regions were converted into gray-scale, and normalized based on the overall mean and variance of the training set.

Text Processing. We first normalize texts by punctuation removal and uppercasing. Similar to Ma et al. (2022), we then used a character-level tokenizer with a vocab size of 41 and 37 for English and Spanish, respectively. In both cases, special symbols, such as the 'space', 'end of sentence', and the 'blank' tokens, were included.

Data Augmentation. We applied SpecAugment (Park et al., 2020) on our audio cues during training with two frequency masks with mask size parameter $F = 27$ and five time masks with adaptive size $pS = 0.05$. Furthermore, we introduce speed perturbation

by randomly selecting a scale factor between 0.9 and 1.1 from a uniform distribution. Regarding video, random cropping of 88×88 pixels, horizontal flipping, and time masking (Ma et al., 2022) with a maximum of 0.4 s were applied during training.

Training Settings. Similar to Burchi and Timofte (2023), we used the Adam optimizer (Kingma and Ba, 2014) and a Noam scheduler (Vaswani et al., 2017) with 10k warmup steps, yielding a peak learning rate of 0.001 with a batch size of 64 samples. All the proposed models were trained for 100 epochs. Those utterances that were longer than 20 s were discarded. In all cases, the loss weight α in Eq. (5) was set to 0.1. Similar to most works in the field, the visual frontend was initialized in all our experiments with the publicly released weights[1] trained on the word-level LRW classification task (Ma et al., 2021a). Additionally, encountering convergence issues with the Spanish video-only system and inspired by the work carried out by Ma et al. (2022), we adopted a cross-lingual learning strategy, initializing both the visual frontend and the encoder backbone with the weights estimated for the English benchmark. Subsequently, we conducted fine-tuning following the training settings outlined in Ma et al. (2022).

Inference Settings. Similar to Ma et al. (2022), we set the LM weight β of Eq. (6) to 0.6 and 0.4 for English and Spanish, respectively. For English, we set a word insertion penalty of 0.5 and a beam size of 40, while for Spanish, we set the word insertion penalty and the beam size to 0.0 and 30, respectively. Regarding the CTC/Attention balance, we set the weight λ of Eq. (6) to 0.1. Note that, during inference, we averaged the model parameters on the 10 epochs with the top validation performance.

Evaluation Metric. Reported results were evaluated by the well-known Word Error Rate (WER) with 95% confidence intervals using the bootstrap method described by Bisani and Ney (2004).

Limitations. Numerous studies (Afouras et al., 2018a; Ma et al., 2021b; Shi et al., 2022; Burchi and Timofte, 2023) investigated the benefits of techniques involving either masking or incorporating noise into acoustic cues during training. While these methods have demonstrated improvements in the robustness of AVSR systems, we consciously refrained from adopting these strategies. Our rationale was to properly study the cross-modality learning capabilities and noise robustness of our proposed method strictly from an architectural design perspective, without incorporating such data augmentation artifacts. Another limitation stems from our computational resource constraints, which made the use of additional large-scale datasets impractical. Nevertheless, our results, achieved without leveraging such datasets, effectively demonstrate the potential of our approach and highlight its focus on data-efficient solutions. Regarding our tailored AVSR architecture, note we also conducted experiments where modalities were forced to use processing branches not ideally suited for them, based on our method. Although performance was worse, the differences were not significant compared to the optimal design, underscoring the need for further exploration of our method's effectiveness, particularly under variable data-scarcity scenarios. This finding also suggests that our model tolerates a degree of architectural flexibility, which is reasonable given that the two Branchformer processing modules were already independently effective for ASR tasks.

5. Results & Discussion[2]

Supporting our Proposed Method. Several reasons motivated and supported the design of our unified, tailored audio–visual encoder. First of all, we trained audio- and video-only systems to corroborate that acoustic and visual cues require different contextual dependencies when recognizing speech. As Fig. 2 reflects, the Branchformer encoder adopted a notably different architecture for each modality, with one of the branches often dominating in almost all layers. We hypothesized that the branch associated with the lower relevance could be pruned without a substantial performance loss. Experimental results reported in Tables 1 and 2 (rows named as "Ours" and "Ours (Tailored)") supported this hypothesis: fine-tuning the tailored version of these modality-specific models for ten epochs yielded recognition rates comparable to their complete counterparts, while reducing the number of parameters by 15.5%. Therefore, we designed the novel tailored audio–visual framework we present in this paper. Albeit the structure of each branch, as Fig. 3 suggests, aligns with the patterns observed in the modality-specific models, note that we also introduced two parameter-shared feed-forward modules to exploit the cross-modality capabilities of our architecture.

Effectiveness of the Tailored AVSR framework. The AVSR task is predominantly assessed in English. For this reason, we first present Table 1 to demonstrate the effectiveness of our proposed parameter-efficient AVSR system, which was able to achieve competitive results compared to the current state of the art. While our approach may be outperformed by various works, they are not directly comparable to our method, since their proposed architectures that not only relied on models comprising vast amounts of parameters, but also on additional large-scale datasets for their pre-training (Liu et al., 2023). Similarly, Table 2 compares our method within the context of the Spanish benchmarks, where we have surpassed the state of the art in most of the datasets by a substantial margin. However, only in the case of MuAViC, our approach exhibits a significant performance drop compared to Burchi et al. (2024) for ASR and AVSR, and Ma et al. (2022), Yeo et al. (2024) for VSR. This disparity can be attributed to several factors, including the larger number of parameters and total training hours in their proposed models. Additionally, a notable difference between these models and our work, as well as other underperforming studies (Anwar et al., 2023; Kim et al., 2023b; Hong et al., 2023), is the use of AVSpeech (Ephrat et al., 2018). The inclusion of this dataset, primarily compiled from TED talks, may have contributed to the performance boost in these state-of-the-art models by providing in-domain data during their pre-training phase.

[1] https://github.com/mpc001/Lipreading_using_Temporal_Convolutional_Networks.

[2] an arxiv version of this article, with the same exact contributions, presents an extended version of our noise and interpretability results discussion and can be found here: https://doi.org/10.48550/arxiv.2407.06606.

Table 1
Comparison of WER (%) to the state of the art on the test sets of the datasets composing the English benchmark for audio-only (A), video-only (V), and audio–visual (AV) settings. Works using non-publicly resources were not considered. Gray shading highlights, along with their corresponding results, those works comparable to ours in terms of either the number of model parameters, hours of training data, or both. Within each subdivision, best results for each dataset are indicated in **bold**.

Method	Modality	No. of Parameters	Total Hours	Additional Datasets	Dataset	
					LRS2-BBC	LRS3-TED
Shi et al. (2022)	A	325M	2192	VoxCeleb2	–	1.5
Ma et al. (2023)	A	250.4M	3448	LRW, VoxCeleb2, AVSpeech	**1.5**	1.0
Burchi et al. (2024)	A	119M	3116	LRW, VoxCeleb2, AVSpeech	–	**0.7**
Ma et al. (2021b)	A	~100M	381/595	LRW	3.9	2.3
Ma et al. (2023)	A	250.4M	818	LRW	2.6	**1.5**
Burchi and Timofte (2023)	A	61.7M	818	LRW	**2.4**	2.0
Ours (Conventional)	A	51.2M	818	LRW	**3.9**$_{\pm 0.5}$	**2.4**$_{\pm 0.4}$
Ours (Tailored)	A	43.3M	818	LRW	4.1$_{\pm 0.5}$	2.6$_{\pm 0.4}$
Shi et al. (2022)	V	325M	2192	VoxCeleb2	–	26.9
Liu et al. (2023)	V	>250M	6720	LRW, VoxCeleb2, AVSpeech, Synthetic	–	**16.9**
Ma et al. (2023)	V	250.4M	3448	LRW, VoxCeleb2, AVSpeech	**14.6**	19.1
Burchi et al. (2024)	V	130M	3116	LRW, VoxCeleb2, AVSpeech	–	25.5
Ma et al. (2021b)	V	~100M	381/595	LRW	37.9	43.3
Prajwal et al. (2022)	V	–	698	–	28.9	40.6
Ma et al. (2022)	V	52M	818	LRW	**27.3**	34.5
Ma et al. (2023)	V	250.4M	818	LRW	27.9	**33.0**
Burchi and Timofte (2023)	V	61.7M	818	LRW	29.8	37.5
Ours (Conventional)	V	60.7M	818	LRW	**29.9**$_{\pm 1.6}$	**37.0**$_{\pm 1.6}$
Ours (Tailored)	V	51.3M	818	LRW	33.6$_{\pm 1.7}$	41.2$_{\pm 1.7}$
Shi et al. (2022)	AV	325M	2192	VoxCeleb2	–	1.3
Ma et al. (2023)	AV	250.4M	3448	LRW, VoxCeleb2, AVSpeech	**1.5**	0.9
Burchi et al. (2024)	AV	197M	3116	LRW, VoxCeleb2, AVSpeech	–	**0.8**
Ma et al. (2021b)	AV	~100M	381/595	LRW	**3.7**	2.3
Burchi and Timofte (2023)	AV	61.7M	818	LRW	2.3	**1.8**
Ours (Conventional)	AV	103.5M	818	LRW	3.0$_{\pm 0.4}$	2.4$_{\pm 0.4}$
Ours (Tailored)	AV	59.3M	818	LRW	**3.0**$_{\pm 0.4}$	**2.1**$_{\pm 0.4}$

Our tailored model stands out as the AVSR architecture with the least parameters, capable of achieving state-of-the-art recognition rates in two different languages, even when trained on a moderate scale of data. Furthermore, given the wide range of data conditions covered by the Spanish benchmark, we demonstrated that our proposed audio–visual framework can effectively address AVSR in diverse domains within a single training procedure.

One interesting finding is that, in both languages, tailored video-only systems showed a significant performance drop compared to their audio-only counterparts, suggesting that VSR is a task more dependent on the model complexity. We also tested the robustness of our AVSR systems in scenarios where acoustic cues are absent and found no significant differences between the complete and tailored architectures.

Noise Robustness. As Table 3 reflects, we evaluated the robustness of our proposed audio-only and audio–visual systems in multiple Signal to Noise Ratio (SNR) levels using babble noise from the NoiseX corpus (Varga and Steeneken, 1993). Consistent with numerous studies (Afouras et al., 2018a; Ma et al., 2021b; Burchi and Timofte, 2023; Anwar et al., 2023), our experiments show that incorporating visual cues significantly enhances the model performance in high-level noise conditions across all studied benchmarks. Our results demonstrate that, despite a reduction in the number of model parameters by nearly half, our proposed tailored audio–visual encoder retained its robustness to noise. For the Spanish benchmark, the benefit of incorporating visual cues, as also seen in Table 2, is less pronounced than for English datasets. This is likely due to the convergence issues observed in video-only Spanish experiments, where the model was not able to achieve acceptable results unless its encoder was pre-trained using weights from the English model. This highlights the need for future work in cross-lingual learning, as discussed above.

Additional preliminary experiments involving the addition noise distortions to acoustic cues during training were also conducted. These data augmentation techniques are expected to increase system robustness to noise and better balance the contributions of audio and visual modalities. While performance under high levels of noise was significantly improved, we did not observe a greater reliance on visual speech cues, with poor results when masking audio input. This finding is also supported by Lin and Harte (Lin and Harte, 2024), who claim that current AVSR systems do not fully exploit the visual modality, highlighting the need for further research in this area.

Interpreting AVSR. In Fig. 2, we can contrast the different patterns observed between the branch weights of an audio-only and a video-only system. Consistent with the findings of the original Branchformer authors (Peng et al., 2022), the audio-only system adopted an architecture where the relevance of each branch changes primarily in an interleaving manner across layers. Conversely, the video-only system exhibits a high dependency on global-context relationships to effectively process the visual speech features. This could be caused by the lack of speech information and the ambiguity inherent to lipreading, forcing the model to rely on

Table 2
Comparison of WER (%) to the state of the art on the Spanish benchmark test sets for audio-only (A), video-only (V), and audio–visual (AV) settings. Works using non-publicly resources were not considered. For each dataset, best results within each modality subdivision are highlighted in **bold**. Cells with missing number of model parameters correspond to approaches of different nature based on HMMs (Gimeno-Gómez and Martínez-Hinarejos, 2022, 2024; Fernandez-Lopez and Sukno, 2022).

Method	Modality	No. of parameters	Total hours	Additional datasets	%WER
			MuAViC$_{es}$		
Anwar et al. (2023)	A	325M	2370	LRW, LRS3, VoxCeleb2	16.5
Burchi et al. (2024)	A	119M	4957	LRW, LRS3, VoxCeleb2, AVSpeech, MuAViC	**7.9**
Ours	A	51.2M	203	–	11.6$_{\pm 0.8}$
Ours (Tailored)	A	43.3M	203	–	12.8$_{\pm 0.8}$
Ma et al. (2022)[a]	V	52M	1546	LRW, LRS2&3, AVSpeech	56.6$_{\pm 0.3}$
Kim et al. (2023b)[a]	V	325M	2825	LRW, LRS3, mTEDx, MLS	70.2
Yeo et al. (2024)	V	250.4M	3832	LRW, LRS2&3, VoxCeleb2, AVSpeech	**45.7**
Ours	V	60.7M	1021	LRW, LRS2&3	77.4$_{\pm 1.5}$
Ours (Tailored)	V	51.3M	1021	LRW, LRS2&3	80.9$_{\pm 1.6}$
Anwar et al. (2023)	AV	325M	2370	LRW, LRS3, VoxCeleb2	15.9
Hong et al. (2023)	AV	>325M	2900	LRW, LRS3, VoxCeleb2, MuAViC	18.0
Burchi et al. (2024)	AV	197M	4957	LRW, LRS3, VoxCeleb2, AVSpeech, MuAViC	**8.2**
Ours (Conventional)	AV	103.5M	361	LRW	11.6$_{\pm 0.7}$
Ours (Tailored)	AV	58.3M	361	LRW	14.3$_{\pm 1.1}$
			CMU-MOSEAS$_{es}$		
Ma et al. (2022)	A	50M	1546	LRW, LRS2&3, AVSpeech	15.4$_{\pm 0.1}$
Ours	A	51.2M	203	–	**12.9**$_{\pm 1.2}$
Ours (Tailored)	A	43.3M	203	–	15.1$_{\pm 1.3}$
Ma et al. (2022)[a]	V	52M	1546	LRW, LRS2&3, AVSpeech	**44.6**$_{\pm 0.6}$
Ours	V	60.7M	1021	LRW, LRS2&3	46.7$_{\pm 2.1}$
Ours (Tailored)	V	51.3M	1021	LRW, LRS2&3	52.6$_{\pm 2.2}$
Ours (Conventional)	AV	103.5M	361	LRW	**13.0**$_{\pm 1.3}$
Ours (Tailored)	AV	58.3M	361	LRW	15.3$_{\pm 1.4}$
			LIP-RTVE		
Ours (Prior Work) (Gimeno-Gómez and Martínez-Hinarejos, 2022)	A	–	9	–	15.3$_{\pm 0.8}$
Ours	A	51.2M	203	–	**7.8**$_{\pm 0.5}$
Ours (Tailored)	A	43.3M	203	–	8.9$_{\pm 0.5}$
Ours (Prior Work) (Gimeno-Gómez and Martínez-Hinarejos, 2022)	V	–	9	–	95.9$_{\pm 0.2}$
Ours	V	60.7M	1021	LRW, LRS2&3	**58.4**$_{\pm 1.0}$
Ours (Tailored)	V	51.3M	1021	LRW, LRS2&3	64.0$_{\pm 1.1}$
Ours (Conventional)	AV	103.5M	361	LRW	**6.7**$_{\pm 0.4}$
Ours (Tailored)	AV	58.3M	361	LRW	7.6$_{\pm 0.5}$
			VLRF		
Ours (Prior Work) (Gimeno-Gómez and Martínez-Hinarejos, 2024)	A	–	1	–	9.1$_{\pm 3.0}$
Ours	A	51.2M	203	–	**4.9**$_{\pm 2.1}$
Ours (Tailored)	A	43.3M	203	–	6.0$_{\pm 2.3}$
Fernandez-Lopez and Sukno (2022)	V	–	–	Synthetic	72.9
Ours (Prior Work) (Gimeno-Gómez and Martínez-Hinarejos, 2024)	V	–	1	–	59.7$_{\pm 4.3}$
Ours	V	60.7M	1021	LRW, LRS2&3	**21.8**$_{\pm 3.0}$
Ours (Tailored)	V	51.3M	1021	LRW, LRS2&3	23.0$_{\pm 3.2}$
Ours (Conventional)	AV	103.5M	361	LRW	**5.0**$_{\pm 1.7}$
Ours (Tailored)	AV	58.3M	361	LRW	5.8$_{\pm 2.2}$

[a] Though results were reported for mTEDx (Salesky et al., 2021), it corresponds to MuAViC, as noted by Anwar et al. (2023).

contextual information to decipher the message. Similar patterns were observed in all the benchmarks studied in our work for video-only systems. However, the audio-only Spanish model did not present that interleaving pattern at the layer level as the English did. This discrepancy between languages in terms of model architecture design might be interesting for the development of multilingual

Table 3
Results in WER (%) of our proposed audio-only (A) and audio–visual (AV) speech recognition systems in babble noisy settings with multiple SNR levels. AV_{conv} and AV_{tail} refer to the conventional and tailored AVSR systems, respectively.

Dataset	Method	SNR levels (dB)				
		–5	0	5	10	15
LRS2-BBC	A	72.8±1.8	17.4±1.5	10.1±0.9	5.0±0.6	4.3±0.6
	AV_{conv}	49.4±1.9	10.8±1.1	6.8±0.7	4.2±0.5	3.6±0.5
	AV_{tail}	45.8±1.9	11.6±1.2	6.0±0.6	3.9±0.5	3.3±0.5
LRS3-TED	A	78.9±1.5	18.4±1.6	9.1±0.7	4.1±0.5	2.7±0.4
	AV_{conv}	62.4±1.7	14.8±1.4	7.1±0.6	3.5±0.4	2.8±0.4
	AV_{tail}	60.5±1.7	13.8±1.3	6.7±0.6	3.6±0.4	2.5±0.4
$MuAViC_{es}$	A	94.5±1.4	34.5±2.2	29.3±1.2	16.8±0.9	13.3±0.8
	AV_{conv}	95.5±1.4	32.9±2.1	28.8±1.2	16.5±0.8	13.1±0.8
	AV_{tail}	92.6±1.2	34.3±2.1	33.3±1.6	20.6±1.2	16.3±1.1
$CMU\text{-}MOSEAS_{es}$	A	>100.0	38.2±3.9	34.9±2.2	19.9±1.5	15.5±1.4
	AV_{conv}	>100.0	38.4±3.5	35.2±2.1	19.0±1.5	14.8±1.3
	AV_{tail}	98.6±1.5	43.5±4.0	37.1±2.2	23.3±1.7	17.6±1.4
LIP-RTVE	A	89.6±1.3	25.6±1.5	18.6±0.8	11.0±0.6	8.8±0.5
	AV_{conv}	90.3±1.3	25.6±1.6	16.3±0.8	9.3±0.6	7.5±0.5
	AV_{tail}	87.4±1.0	26.9±1.6	18.2±0.8	10.8±0.6	8.6±0.5
VLRF	A	>100.0	57.0±14.2	34.5±6.0	12.8±3.5	7.3±2.3
	AV_{conv}	>100.0	82.4±15.9	45.8±8.0	12.6±3.0	6.2±2.1
	AV_{tail}	>100.0	59.3±13.1	35.4±6.4	12.1±3.0	6.6±2.5

ASR systems (Anwar et al., 2023; Hong et al., 2023), paving the way for future research lines. Regarding our AVSR systems, we also inspected the relative weight learned during training when fusing the audio and visual cues. In all cases, regardless of the language or the architecture, our adaptive fusion module consistently assigned roughly 73.1±12.3% relevance to the acoustic cues. This observation aligns with the research of Duchnowski et al. (2000), which supported that only around 30% of speech information is visible. Additional experiments were carried out to investigate the variability of the learnable relative weights based on the sample length and noise degradation in audio-only, video-only, and audio–visual settings. Nevertheless, in line with the findings reported by the original authors (Peng et al., 2022), the weights exhibited remarkable stability across diverse conditions. Our analysis provides initial insights into the architectural differences, beyond the frontends, when dealing with speech features from different modalities and how relevant each of them can be when decoding speech.

6. Conclusions & Future work

In this paper, we introduced a novel framework for the tailored design of unified audio–visual encoders in the context of parameter-efficient AVSR by leveraging the interpretability and flexibility offered by the Branchformer architecture (Peng et al., 2022). Through extensive experiments across multiple languages and data conditions, we showed the effectiveness of our proposed encoder, which was able to achieve competitive results with a moderate amount of training data, while significantly reducing the number of parameters composing the model. Regarding our future work, we intend to focus our efforts towards developing lightweight AVSR systems for real-world applications. This includes exploring the combination of our tailored audio–visual encoder with parameter-sharing techniques, such as the Sim-T architecture (Wei et al., 2023); the integration of audio–visual cues in early stages through, for instance, the implementation of intermediate auxiliary losses (Ma et al., 2022; Lee and Watanabe, 2021); and finally the shift to non-autoregressive paradigms, such as the Mask-CTC decoder (Higuchi et al., 2021), capable of achieving state-of-the-art recognition rates in much less inference time. In this context, we further aim to investigate the potential of cross-modal and cross-lingual learning within parameter-efficient AVSR frameworks.

CRediT authorship contribution statement

David Gimeno-Gómez: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Carlos D. Martínez-Hinarejos:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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